

AdaptiveMaxPool3d#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.pooling.AdaptiveMaxPool3d.html>

Description:

Applies a 3D adaptive max pooling over an input signal composed of several input planes. The output is of size $D_{out} \times H_{out} \times W_{out}$, for any input size. The number of output features is equal to the number of input planes.

`output_size` (Union[int, None, tuple[Optional[int], Optional[int], Optional[int]]]) – the target output size of the image of the form $D_{out} \times H_{out} \times W_{out}$. Can be a tuple $(D_{out}, H_{out}, W_{out})$ or a single D_{out} for a cube $D_{out} \times D_{out} \times D_{out}$. D_{out} , H_{out} and W_{out} can be either a int, or None which means the size will be the same as that of the input.

`return_indices` (bool) – if True, will return the indices along with the outputs. Useful to pass to `nn.MaxUnpool3d`. Default: False

Input: $(N, C, D_{in}, H_{in}, W_{in})$ or $(C, D_{in}, H_{in}, W_{in})$

Function Signature

```
class  
torch.nn.modules.pooling.AdaptiveMaxPool3d(output_size,  
return_indices=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Code Examples

```
>>> # target output size of 5x7x9  
>>> m = nn.AdaptiveMaxPool3d((5, 7, 9))  
>>> input = torch.randn(1, 64, 8, 9, 10)  
>>> output = m(input)  
>>> # target output size of 7x7x7 (cube)  
>>> m = nn.AdaptiveMaxPool3d(7)  
>>> input = torch.randn(1, 64, 10, 9, 8)  
>>> output = m(input)  
>>> # target output size of 7x9x8  
>>> m = nn.AdaptiveMaxPool3d((7, None, None))  
>>> input = torch.randn(1, 64, 10, 9, 8)  
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies a 3D adaptive max pooling over an input signal composed of several input planes.

The output is of size $D_{out} \times H_{out} \times W_{out}$ for any input size. The number of output features is equal to the number of input planes.

`output_size` (Union[int, None, tuple[Optional[int], Optional[int], Optional[int]]]) – the target output size of the image of the form $D_{out} \times H_{out} \times W_{out}$. Can be a tuple (Dout,Hout,Wout) or a single Dout for a cube $D_{out} \times D_{out} \times D_{out}$. Dout, Hout and Wout can be either a int, or None which means the size will be the same as that of the input.

`return_indices` (bool) – if True, will return the indices along with the outputs. Useful to pass to `nn.MaxUnpool3d`. Default: False

Input: (N,C,Din,Hin,Win) or (N, C, D_{in}, H_{in}, W_{in}) or (C,Din,Hin,Win) or (C, D_{in}, H_{in}, W_{in}).

Output: (N,C,Dout,Hout,Wout) or (N, C, D_{out}, H_{out}, W_{out}) or (C,Dout,Hout,Wout), where (Dout,Hout,Wout)=`output_size`(D_{out}, H_{out}, W_{out})=text{output_size}(Dout,Hout,Wout)=`output_size`.

Runs the forward pass.